

Agentic Commerce Governance Framework™: Classifying Enterprise Liability Exposure in the Age of Machine Buyers

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Abstract

The emergence of autonomous AI agents as active participants in commercial transactions represents the most significant structural shift in enterprise architecture since the advent of e-commerce. This paper introduces the Agentic Commerce Governance Framework™, a six-level classification system (Level 0 through Level 5) that maps the progressive elimination of human psychological friction from the purchase transaction. The framework identifies Level 3 (Delegated Execution) as the threshold at which traditional risk governance models first break down, and defines three structural governance failures that create liability exposure at Levels 3 through 5: liability without visibility, the breakdown of psychological friction as a control mechanism, and data maturity deficits that transform product catalogs into arbitrage vectors. The paper introduces a dual governance architecture combining the Mandate Protocol for top-down agent authorization with tactical exposure diagnostics for bottom-up vulnerability identification as a framework for establishing comprehensive organizational governance posture in advance of agentic commerce deployment.

Keywords: agentic commerce, AI governance, autonomous agents, enterprise architecture, liability exposure, API governance, machine buyers, digital transformation

1. Introduction

1.1 Arrival of the Machine Buyer

For two decades, enterprise digital commerce infrastructure was built around a single assumption: the buyer is human. Human buyers respond to brand heritage, visual design, loyalty incentives, and interface friction. They hesitate. They abandon carts. They respond to urgency cues. Every system designed to convert, retain, and upsell was calibrated to the psychological architecture of the human decision-maker.

That assumption is no longer structurally valid.

Autonomous AI agents are now operating as active commercial participants. They interpret complex, multi-variable purchase objectives. They navigate APIs directly, bypassing visual interfaces entirely. They evaluate inventory, calculate tax liability, apply loyalty balances, and execute payment without a human present at any step of the transaction. The buyer, in an increasing number of commercial interactions, is a machine.

Google Marketing Live 2026 (May 20, 2026) marked a watershed moment in this transition. Google announced the live deployment of the Universal Commerce Protocol, co-developed with Shopify, Etsy, Wayfair, Target, and Walmart, with Amazon, Meta, Microsoft, Salesforce, and Stripe as steering participants, covering retail, hotel booking, and local food delivery. Google simultaneously launched conversational feed attributes specifically designed for AI surfaces, loyalty authentication wired directly into autonomous checkout, and a Share of Voice metric for AI-driven search presence. These are not projections. They are live infrastructure.

The machine buyer is not arriving. It has arrived. McKinsey & Company projects that AI agents could mediate between \$3 trillion and \$5 trillion of global consumer commerce by 2030, with \$1 trillion in US B2C retail alone (Schumacher et al., 2025).

1.2 Governance Problem

The enterprise governance response to this transition has been structurally inadequate. Organizations have focused primarily on the technical dimensions of AI deployment while failing to address the organizational and legal dimensions that autonomous agent behavior creates.

The core governance problem is not technical. It is organizational. When an autonomous agent executes a commercial transaction, three questions must be answerable: Who authorized this agent to act? Who is accountable when it acts incorrectly? How can it be stopped when necessary?

In the majority of organizations that have deployed or are actively deploying AI agents, these questions have no definitive answers. Authorization structures have not been formalized. Accountability has not been assigned. Halt mechanisms have not been designed. The International Monetary Fund has identified this as a systemic risk, noting that traditional authorization mechanisms, including Know Your Customer protocols and multi-factor authentication, are designed for human users who explicitly approve

transactions, and that when AI agents initiate payments, verifying the identity of the agent and the intent of the user becomes significantly more complex (Davidovic & Tourpe, 2026).

This paper argues that enterprise liability exposure in agentic commerce is a function not of technological sophistication but of organizational readiness, and that the level of autonomous agent operation an organization deploys must be matched by an equivalent level of governance maturity.

1.3 Research Context and Framework Origin

The Agentic Commerce Governance Framework™ was developed through a combination of field observation across enterprise transformation engagements spanning 28 years and 30 countries, and analytical design in response to the specific governance challenges emerging from AI agent deployment in commercial contexts. The framework was subsequently deployed as the diagnostic backbone of the Agentic Exposure Audit, a structured architectural assessment instrument for enterprise boards and technical leadership (Meltke, 2026).

The six-level classification system at the framework's core was derived inductively from observed patterns of agent behavior in operational commercial environments, then refined through analytical examination of the governance thresholds at which existing organizational structures first fail to provide adequate oversight.

2. Theoretical Foundation

2.1 Architecture of Human-Centered Commerce

The dominant paradigm of digital commerce governance rests on an implicit architectural assumption: that a human decision-maker is present at critical transaction points. This assumption is embedded not merely in UX design but in the legal, compliance, and risk frameworks that govern commercial transactions.

Fraud detection algorithms are calibrated to human behavioral patterns including session timing, navigation paths, and hesitation signals. Chargebacks are governed by consumer protection frameworks that assume a human initiated the purchase. Dynamic pricing policies are designed around the expectation that a human will evaluate the displayed price before proceeding. Every layer of this architecture was designed for a buyer with psychological friction: a decision-maker who can be influenced, who can hesitate, who can change their mind, and who can be held legally accountable.

The machine buyer does not have psychological friction. It does not respond to brand heritage or urgency cues. It does not hesitate. It processes data against objective criteria and executes decisions at machine speed across multiple simultaneous sessions. When an AI agent encounters a pricing feed or inventory API designed for human-speed browsing, it does not engage with that architecture as its designers intended. It exploits it.

2.2 Friction Elimination Progression

The transition from human-centered to machine-centered commerce does not occur at a single point. It occurs progressively, as autonomous agent capability advances and human presence in the transaction is systematically reduced.

The critical insight driving the Agentic Commerce Governance Framework™ is that this progression is not merely a technical continuum. It is a governance discontinuity. Each level of increased agent autonomy eliminates a category of organizational control that existing governance structures assume is present. The governance failure does not compound gradually: it cascades at specific thresholds where the removal of human friction eliminates an entire control mechanism simultaneously.

The framework maps this progression across six levels, from pure manual commerce at Level 0 to fully autonomous closed-loop operation at Level 5. The governance crisis does not begin at Level 5. It begins at Level 3.

3. The Agentic Commerce Governance Framework™

3.1 Framework Overview

The Agentic Commerce Governance Framework™ classifies autonomous agent involvement in commercial transactions across six levels defined by two primary variables: the degree of human psychological friction remaining in the transaction, and the point at which the governance structures of a standard enterprise organization first fail to provide adequate oversight.

Table 1 presents the framework overview. Levels 0 through 2 represent the zone of traditional governance validity. Levels 3 through 5 represent the governance crisis zone, where the removal of human decision-making at final transaction points eliminates control mechanisms that organizational governance structures have not been redesigned to replace.

Table 1: Agentic Commerce Governance Framework — Classification Overview

Level	Classification	Friction State	Governance Crisis
0	Pure Manual Commerce	Maximum	None
1	Intent Matching & Discovery	High	None
2	Assisted Data Routing	Reduced	Emerging
3	Delegated Execution	Low	Begins here
4	Advanced Buy	Near zero	Acute
5	Closed Loop Autonomy	Absolute zero	Full exposure

Source: Meltke, F. (2026). Agentic Commerce Governance Framework™. contraco Management Consulting.

3.2 Level 0: Pure Manual Commerce

At Level 0, AI plays no role in the transaction. Human buyers use traditional search engines, navigate individual merchant websites, manually evaluate inventory, and execute payment. Every point of discovery, data entry, and authentication rests entirely on the human operator. Governance structures at Level 0 are fully adequate.

3.3 Level 1: Intent Matching and Discovery

At Level 1, the agent interprets complex multi-variable human intent and surfaces relevant matching products. The moment the human clicks a link, the agent's involvement terminates. The agent functions as an enhanced search instrument, not a transactional actor. Governance structures remain fully adequate.

3.4 Level 2: Assisted Data Routing

At Level 2, the agent navigates to the checkout page, pre-populates shipping addresses, inputs billing details, and applies relevant loyalty codes via browser automation or API integration. It halts at the final checkpoint: the human must physically confirm submission. Governance structures are substantially adequate, with emerging exposure in data handling and pre-population logic that merits monitoring.

3.5 Level 3: Delegated Execution — The Governance Crisis Threshold

Level 3 is the governance threshold at which traditional risk assessment models first break down. At this level, the human grants the agent explicit, time-bound permission to execute a specific purchase within defined boundaries. The agent holds secure access to tokenized payment credentials and executes the final API handshake autonomously. If any variable falls outside the mandate, the transaction aborts.

The governance crisis at Level 3 is not that the agent acts autonomously. It is that the authorization structures required to govern autonomous action are typically absent. Three structural governance failures become acute at Level 3 and compound through Levels 4 and 5, as analyzed in Section 4.

3.6 Level 4: Advanced Buy — The API Arbitrage Zone

At Level 4, the agent processes broad, outcome-based objectives. It orchestrates multi-merchant transactions inside a single virtualized environment, evaluates live inventory, calculates regional tax liabilities, and executes separate downstream payments simultaneously. Google's Universal Commerce Protocol and Universal Cart, announced at Google Marketing Live 2026, represent the first large-scale public deployment of Level 4 infrastructure (Google, 2026).

Level 4 is no longer theoretical. Any enterprise with a transactional API surface exposed to Google's ecosystem is architecturally operating in a Level 4 environment, regardless of whether its internal governance structures have been designed for that exposure.

3.7 Level 5: Closed-Loop Autonomy — Full Liability Exposure

At Level 5, the agent operates continuously without requiring human prompts. It monitors inventory depletion, predicts demand, negotiates machine-to-machine pricing with merchant agents, issues smart contract micro-payments, and handles automated returns without human notification unless policy thresholds are exceeded. Commerce becomes an invisible, self-sustaining utility layer.

At Level 5, the absence of governance infrastructure does not merely create exposure. It constitutes complete liability transfer to an organizational layer that was never designed to carry it.

4. Three Structural Governance Failures

Analysis of enterprise AI deployments across the governance crisis zone (Levels 3 through 5) reveals three structural failures that consistently account for the governance gap between technological capability and organizational readiness. Table 2 summarizes the three failure types.

Table 2: Three Structural Governance Failures in Agentic Commerce

Failure Type	Organizational Manifestation
Liability without Visibility	Legal and compliance frameworks assign transaction liability to departments that have no oversight of the autonomous agent layer executing those transactions.
Breakdown of Psychological Friction	Traditional enterprise architecture was designed for human buyers influenced by brand cues and visual UI. Agents bypass these elements entirely, operating on algorithmic physics at the API layer.
Data Maturity Deficits	Enterprise catalogs treat product data as passive content rather than machine-legible infrastructure. Unverified API endpoints become arbitrage vectors exploitable at machine speed, often without organizational intent or consent.

4.1 Liability without Visibility

The most consequential structural failure in agentic commerce governance is the systematic disconnect between where legal liability is formally assigned and where operational awareness of agent behavior actually resides. Legal and compliance frameworks assign transaction liability to departments that have no operational oversight of the autonomous agent layer executing those transactions. These departments are accountable for outcomes they cannot observe, using frameworks designed for human-initiated transactions they can audit after the fact.

The AI agent is operated by technical teams that have full visibility into agent behavior but carry no formal liability for commercial outcomes. The organization has structurally separated accountability from visibility. The European Union's AI Act (European Commission, 2024) imposes provider and deployer accountability obligations that presuppose governance structures most organizations have not yet built.

When an autonomous agent executes an unintended transaction, the organization faces liability for outcomes that no internal role was positioned to observe, prevent, or halt.

4.2 The Breakdown of Psychological Friction as a Control Mechanism

Enterprise architecture and risk governance were not designed with autonomous agents in mind. They were designed for human buyers, and they depend, often without explicit

acknowledgment, on psychological friction as a control mechanism. Human buyers hesitate. They read terms and conditions selectively. They respond to warnings and confirmation dialogs. In aggregate, this psychological friction functions as a distributed control system that enterprise risk models have implicitly relied upon.

AI agents operating at Levels 3 through 5 do not experience psychological friction. They process pure data against objective criteria. They do not respond to warning dialogs or confirmation prompts. The IMF identifies this as a fundamental tension between the probabilistic nature of AI behavior and the deterministic requirements of payment infrastructure (Davidovic & Tourpe, 2026). When psychological friction is eliminated from the transaction, every control mechanism that depended on it fails simultaneously.

4.3 Data Maturity Deficits as Arbitrage Vectors

The third structural failure is the treatment of enterprise product data as passive content rather than machine-legible infrastructure subject to active governance. In the context of agentic commerce, data maturity is defined not by the accuracy of the data for human consumption, but by the machine-legible structure and security posture of the API endpoints hosting that data. An enterprise catalog designed for human browsing presents products in a format optimized for visual processing. An autonomous agent consuming the same catalog through an API processes it as raw data. Pricing structures designed to anchor human perception become optimization inputs for algorithmic purchasing strategies.

API endpoints designed for human-speed browsing, without agent-specific authentication or rate governance, become arbitrage vectors exploitable across thousands of simultaneous sessions. Google's launch of conversational feed attributes at Google Marketing Live 2026 formalized this reality: products without properly structured machine-legible attributes are structurally invisible to autonomous purchasing agents on conversational queries (Google, 2026).

5. The Dual Governance Architecture

Addressing the three structural governance failures described in Section 4 requires a dual-architecture approach. A single governance instrument cannot simultaneously provide the top-down organizational alignment needed to define and control agent authorization, and the bottom-up technical visibility needed to identify specific infrastructure vulnerabilities.

The framework proposes two complementary instruments: the Mandate Protocol for top-down governance, and the Exposure Diagnostic for bottom-up vulnerability identification. Together, they establish what the framework terms comprehensive governance posture, an organizational state in which both strategic intent and operational reality are addressed before autonomous agent deployment.

5.1 The Mandate Protocol: Top-Down Authorization Governance

The Mandate Protocol is a structured organizational process for defining, documenting, and monitoring the authorized scope of AI agent action before deployment. Its value lies not in the technical specifications it produces but in the organizational process of producing them, a process that consistently surfaces governance assumptions that were never made explicit and accountability gaps that were never recognized.

The Protocol requires four organizational decisions for each deployed agent, as detailed in Table 3.

Table 3: Mandate Protocol — Required Organizational Decisions

Protocol Element	Organizational Decision Required
Authorization	Which human role formally approved this agent's deployment and scope of action, and what accountability does that role carry for agent behavior?
Scope Definition	What specific actions is this agent authorized to take, under what conditions, and with what explicit exclusions?
Escalation Design	Under what conditions must the agent stop and request human decision-making, and which human role receives that escalation?
Audit Architecture	How is agent behavior logged, what is the retention period, and who reviews it and at what frequency?

The Mandate Protocol is deliberately designed as an organizational governance document that precedes technical configuration. Authorization, scope, escalation, and audit decisions must be made at the organizational level by the human roles that carry accountability for agent behavior, before those decisions are encoded in technical parameters. The Protocol directly addresses the liability-visibility disconnect identified in Section 4.1 by requiring formal authorization decisions from accountable human roles and establishing explicit escalation paths and audit review responsibilities.

5.2 The Exposure Diagnostic: Bottom-Up Vulnerability Identification

The Mandate Protocol establishes who is accountable and what is authorized. The Exposure Diagnostic identifies where the architecture is currently vulnerable. The two instruments are designed to operate in sequence: the Mandate Protocol defines the governance intent; the Exposure Diagnostic maps the gap between that intent and the technical reality.

The Exposure Diagnostic is structured around three questions that function as primary vulnerability indicators:

- What is the current maturity level of machine-legible data feeds? This question surfaces the data maturity deficit described in Section 4.3. Most organizations discover that their product data infrastructure has been designed for human consumption and lacks the structured attributes, agent-specific access controls, and rate governance required for secure machine consumption.
- How does infrastructure currently distinguish between a human user and an authorized AI proxy? This question surfaces the authentication gap that is the primary technical enabler of the arbitrage exposure described in Section 4.3. API endpoints that cannot cryptographically distinguish an authorized agent expose pricing and inventory data to machine-speed exploitation.
- Which department holds legal liability for autonomous transactions in the strategic roadmap? This question surfaces the liability-visibility disconnect described in Section 4.1. In most organizations, this question has no single definitive answer, or produces answers from multiple departments that contradict each other.

5.3 The Governance Maturity Matching Principle

The central governance principle emerging from the dual-architecture framework is the Governance Maturity Matching Principle: the level of autonomous agent operation an organization deploys must be matched by an equivalent level of governance maturity.

This principle has an important practical implication. The framework does not argue that organizations should avoid higher-level agent deployment. Level 4 infrastructure is already live in the commercial environments of any enterprise participating in Google's ecosystem. The principle argues instead that governance investment must scale with agent autonomy. An organization deploying at Level 3 must have Mandate Protocol authorization in place. An organization exposed to Level 4 infrastructure must have machine-legible data governance and agent authentication architecture in place. An organization planning Level 5 operations must have closed-loop audit architecture and continuous monitoring in place.

6. Empirical Observations

The framework presented in this paper is grounded in field observation across enterprise AI and transformation engagements conducted under non-disclosure agreements. While the NDA constraints governing the underlying research prevent publication of identifying organizational data, the structural patterns that the framework addresses are consistently observable across sectors and geographies.

Four empirical observations merit explicit statement, as they directly inform the framework's design.

- Organizations consistently overestimate their data maturity for agentic commerce contexts. Internal assessments conducted using standards designed for human-interface digital commerce systematically fail to identify the structural gaps in machine-legible data architecture that become consequential when agents are the primary consumers of that data.
- The liability-visibility disconnect is the most consistent finding across governance assessments. When organizations are asked to identify which, human role holds accountability for an autonomous agent's commercial actions, the process of answering the question consistently surfaces previously unrecognized gaps.
- The absence of agent authentication infrastructure is near-universal in organizations that have not conducted a dedicated agentic architecture review. API endpoints designed before the emergence of Level 3 and above agent deployment were not designed to distinguish agent identity, and retrofitting authentication into production infrastructure is consistently more complex than organizations anticipate.
- Organizations that complete the Mandate Protocol before deployment consistently report lower governance incident rates than organizations that deploy without formal mandate definition. The structured process of answering the Protocol's four questions surfaces organizational misalignments that, left unaddressed, produce authorization disputes and accountability gaps when incidents occur.

These observations are presented as practitioner findings subject to academic validation through independent research. The framework is offered as a contribution to the practitioner and research community for testing, refinement, and extension.

7. Limitations and Future Research

The framework presented in this paper has three primary limitations that academic and practitioner researchers should consider in applying and extending it.

First, the observational base is drawn from contraco's client engagements and therefore represents organizations that had already decided to engage external transformation support. Organizations managing agentic commerce governance entirely through internal resources may exhibit different structural patterns. The framework's applicability to these organizations is asserted but not empirically validated.

Second, the six-level classification system, while derived from observed patterns, is a simplification of a complex continuum. Actual enterprise deployments frequently exhibit characteristics of multiple levels simultaneously. The framework's treatment of levels as discrete classifications is analytically useful but should not be mistaken for an empirical claim that organizations operate cleanly within single levels.

Third, the framework was developed primarily from commercial transaction contexts. Its applicability to procurement, supply chain, and B2B agentic deployments, where the governance dynamics are structurally similar, but the organizational and legal contexts differ, has not been fully tested.

Future research directions include longitudinal studies of governance incident rates in organizations that have implemented Mandate Protocol governance compared to those that have not; cross-sector analysis of data maturity assessment methodologies specific to agentic commerce contexts; investigation of legal and regulatory framework adaptation in jurisdictions where AI Act compliance intersects with agentic commerce deployment; and empirical measurement of the threshold at which Level 4 infrastructure exposure requires Level 4 governance maturity to prevent material commercial harm.

8. Conclusion

The machine buyer has arrived. The commercial infrastructure to support autonomous agent transactions at scale is live and expanding. The governance infrastructure to manage the liability, accountability, and control challenges that autonomous agent transactions create is, in the majority of enterprises, absent.

The Agentic Commerce Governance Framework™ offers a classification system for understanding where an organization's current and planned agent deployment sits on the spectrum from human-centered to machine-centered commerce, and for identifying the governance threshold at which existing organizational structures first fail to provide adequate oversight.

The governance crisis in agentic commerce is a crisis of sequencing. Organizations have deployed agent capability without deploying equivalent governance capability. Closing that gap requires deliberate design: formal authorization structures, explicit accountability assignment, and infrastructure that treats agent security as a first-order requirement. Organizations seeking to formalize this posture can initiate an Agentic Exposure Audit to establish their baseline capability.

Your architecture either governs the autonomous economy, or it is governed by it.

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